ETH zürich

Mobile Computing

Socio-Economic and Humanitarian Use

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

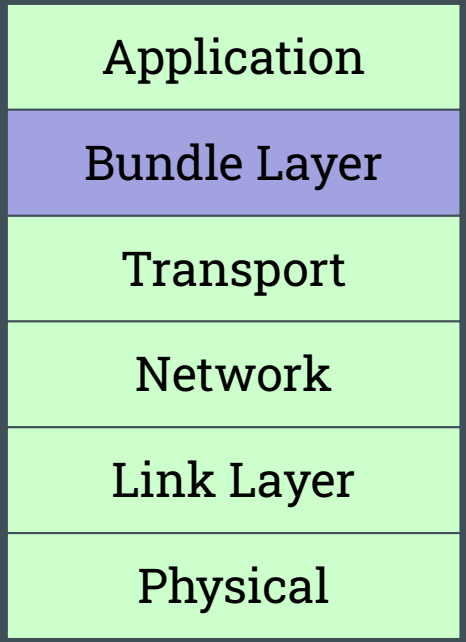
Tournament

Delay Tolerant Networking, DTN

- The DTN idea originated from the Interplanetary Networking Special Interest Group (IPNSIG)
  - Deep space networks and spacecraft communications
- Other use cases:
  - Networks with intermittent connectivity (remote areas)
  - Military networks
- Can tolerate delay or disruption, or disconnection
- Data travels along with moving devices

DTN Layer in the Protocol Stack

- Creates time-stamped bundles (data units)
  - Open for different DTN routing schemes
  - Supports TCP and UDP convergence layer
- IETF Standards
  - rfc 4838: Delay-Tolerant Network Architecture
  - rfc 5050: Bundle Protocol Specification
- Challenges
  - Unpredictable link error rates, very large delays
  - Unpredictable network capacity: Low with a small number of devices, but high with a large number of devices



Case 1: Project with ICRC ...

ETH4D Humanitarian Action Challenges (HAC)

- Wireless Mesh Networking for Humanitarian Communication
- Mesh and DTN to deliver data in challenged environments

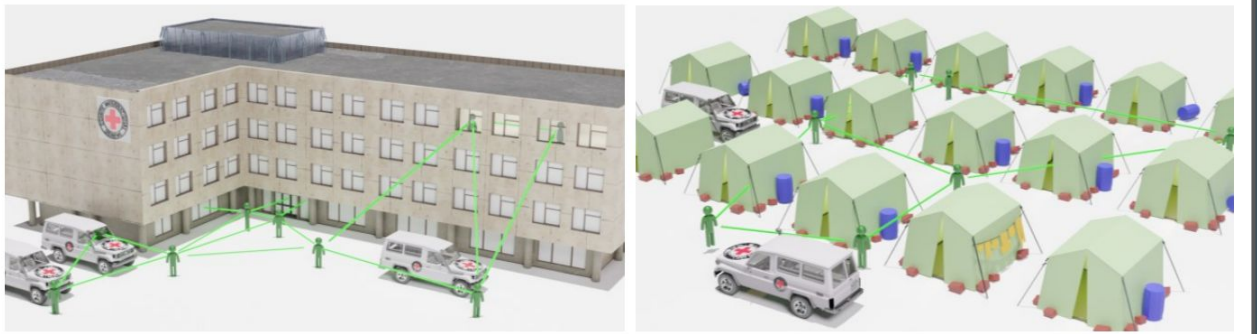
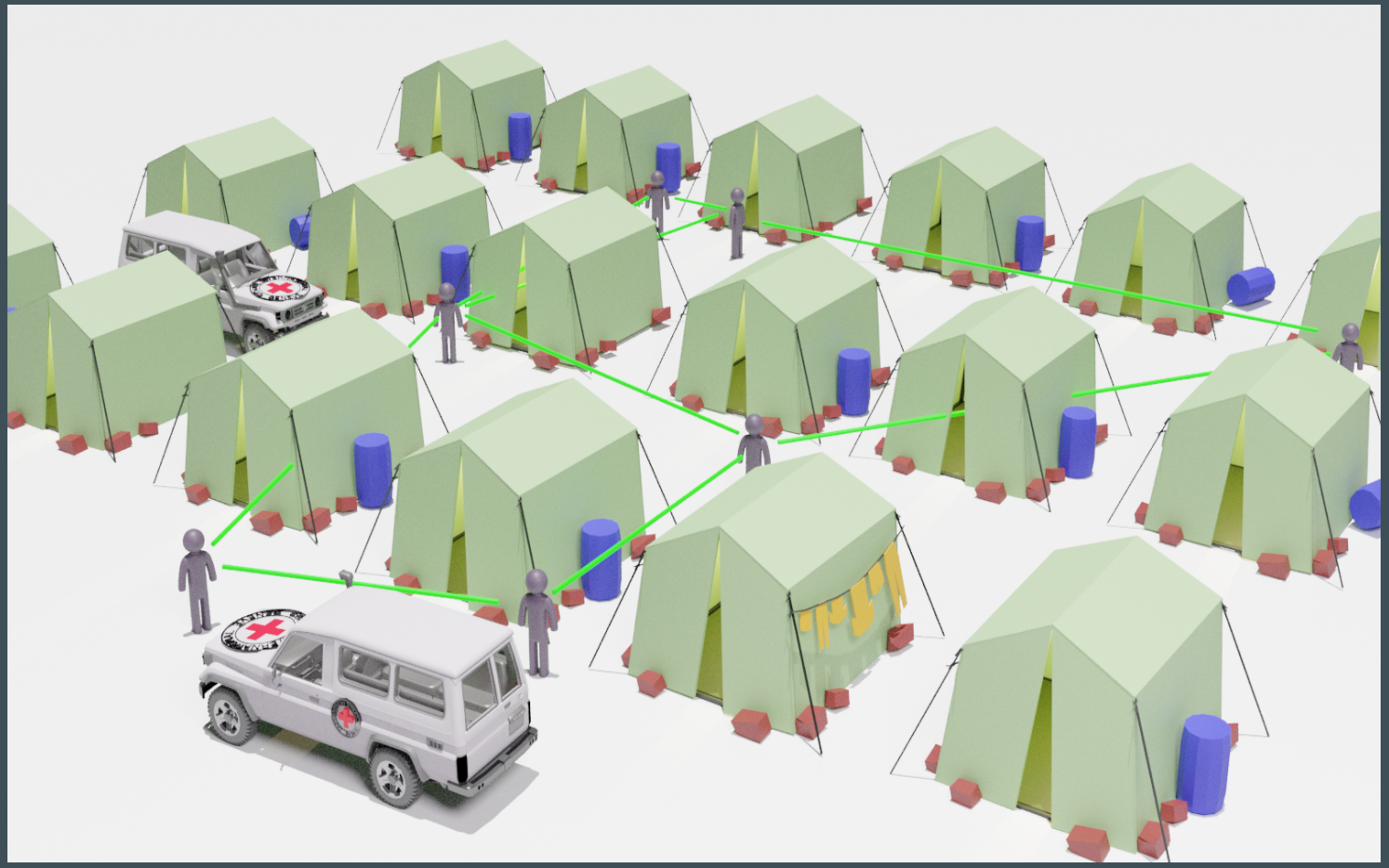


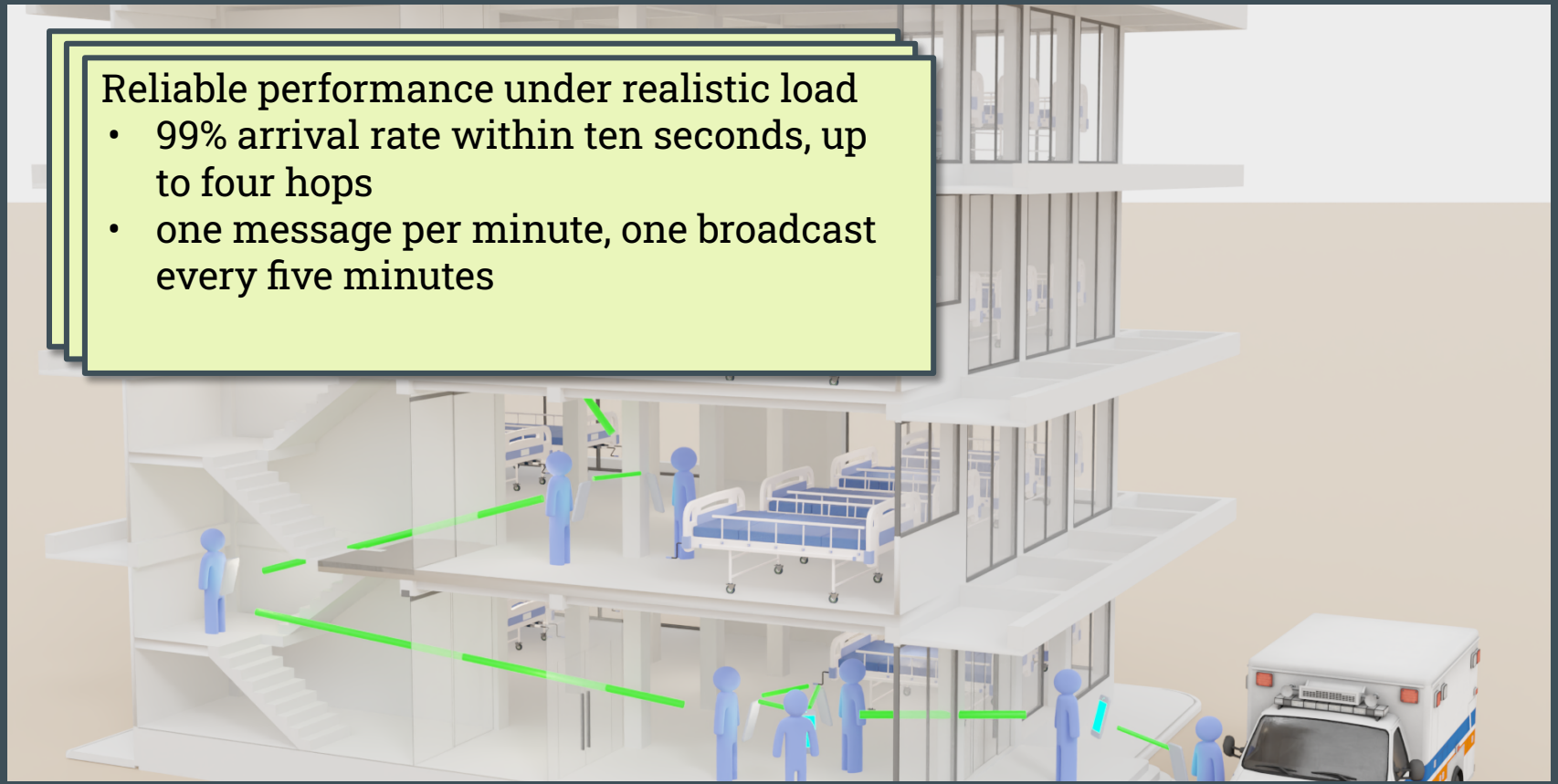
Figure 1: Humanitarian workers remain connected with each other through wireless mesh networks, when cellular networks and non-terrestrial (satellite) networks (NTNs) are not available. Some devices could very well be connected through satellite to provide all other devices with access to the wide area network (public internet). LEFT: Hospital scenario with outdoor and indoor use; RIGHT: Tent camp with outdoor use.



Wireless Mesh Networking

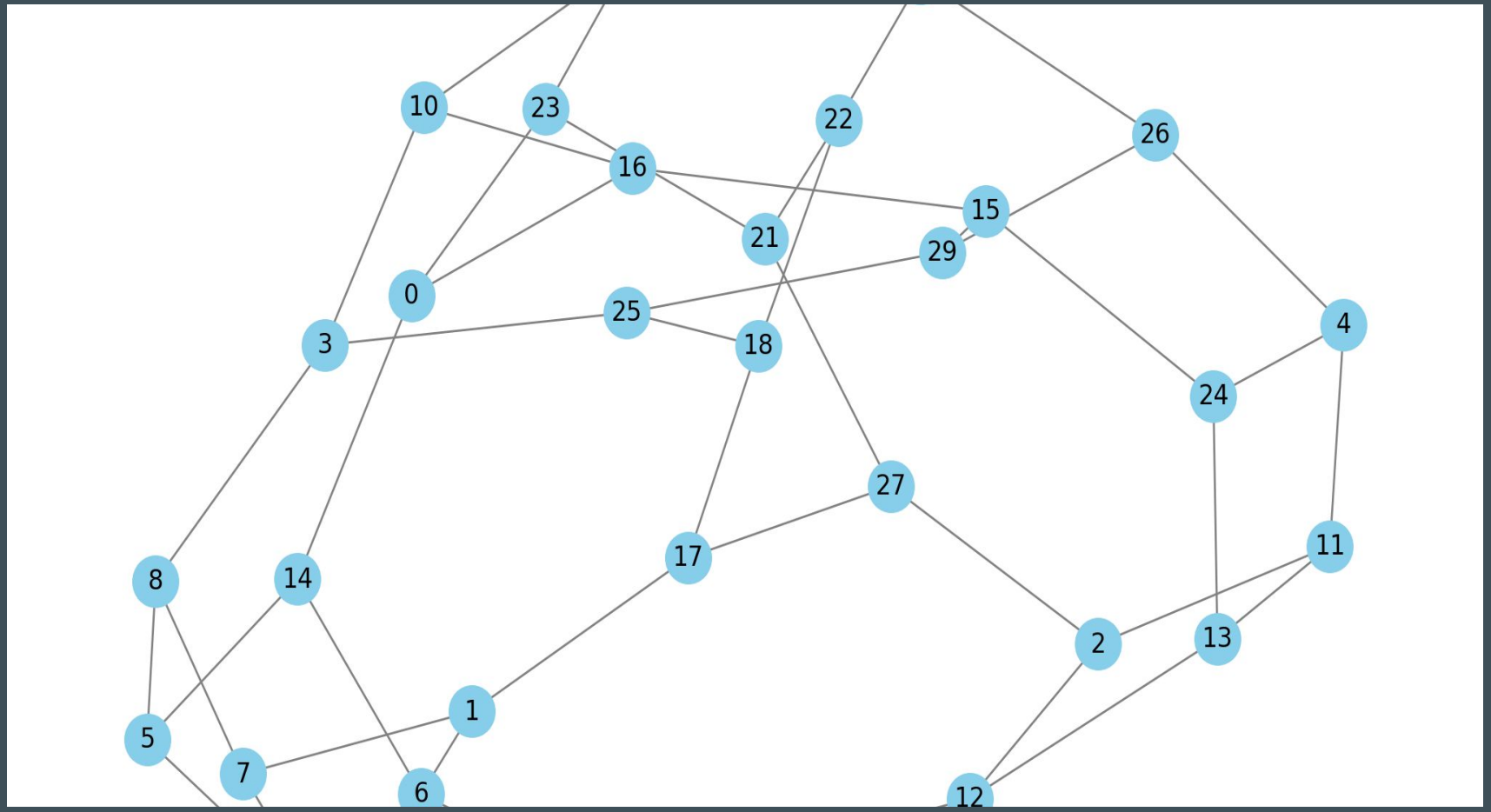
- Goal: A mobile app to create a network of mobile devices, operating without any additional telecom infrastructure
- Improve humanitarian workers communication during emergencies
- Adhere to ICRC’s security and privacy constraints
- General relevance for the humanitarian sector in disaster relief (earthquakes, power outages)
- Duration: Sep 2024 … Feb 2026

Wireless Mesh Networking

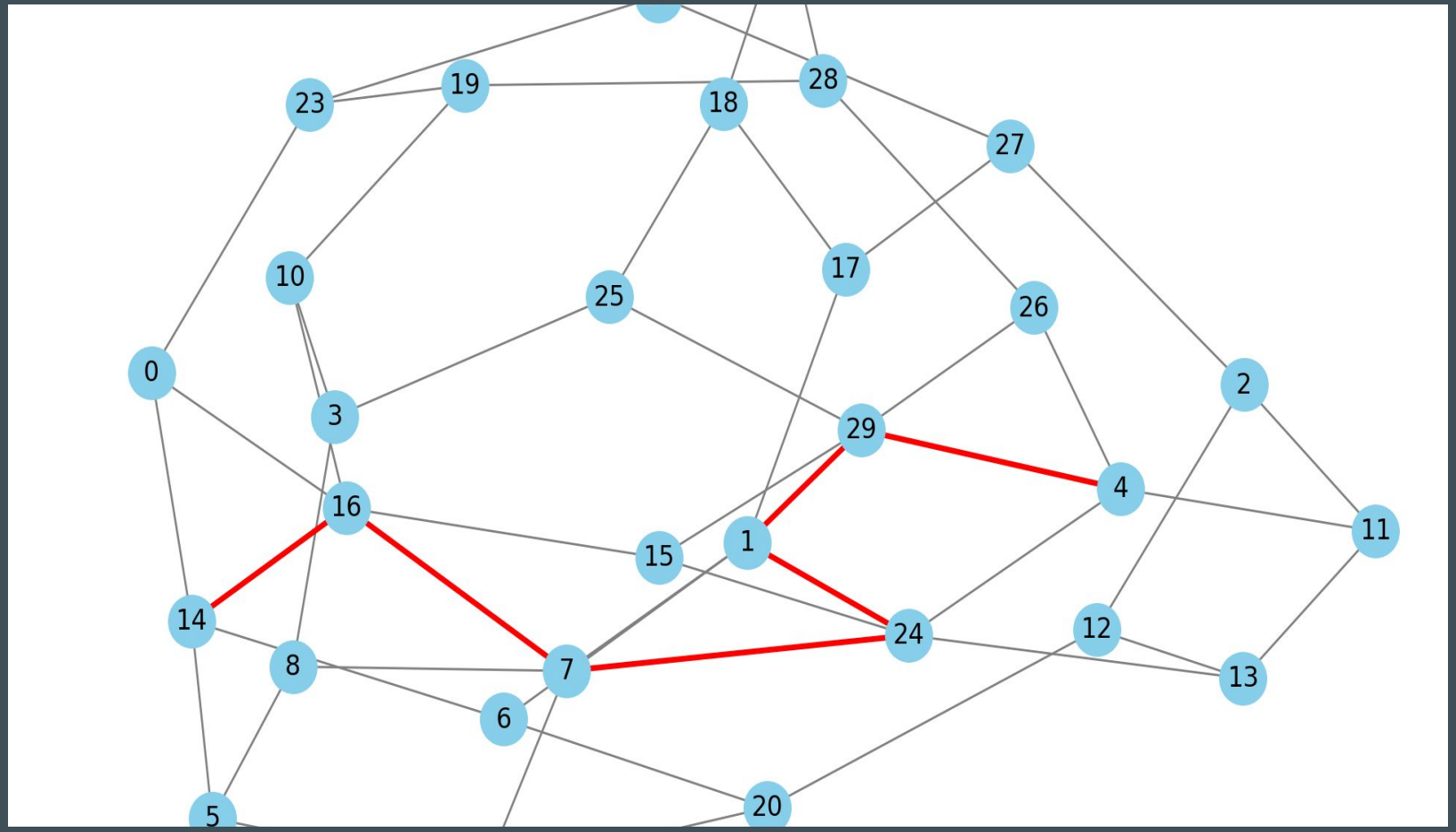


Reset
RedMesh
Network demonstration: This is an experimental prototype developed for internal testing and demonstration purposes, focussing on wireless mesh networking. The recommended use is for evaluation and feedback only.
Let's go!
Reset
RedMesh Demo
Discovered Peers ("⭐" if connected)
(no devices discovered)
Messages
...
...
...
...
Demo
About RedMesh
Settings

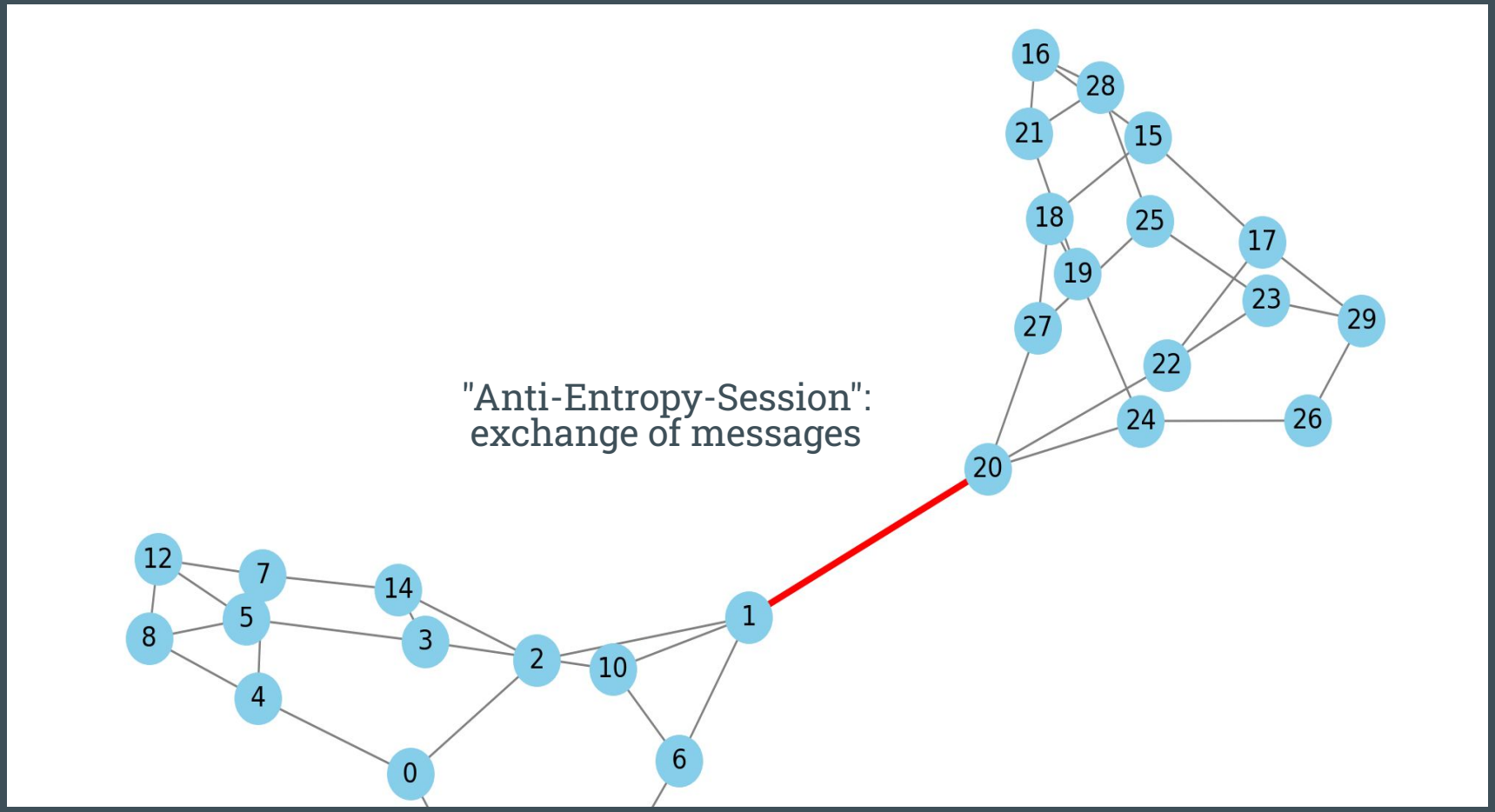
Layer 2 – Wi-Fi Aware Mesh Networking – Native Implementation on Android and (soon) on iOS



Layer 3 – Routing with Link State Announcements – React Implementation



Layer 4 – Delay Tolerant Networking (Epidemic Routing) - React Implementation

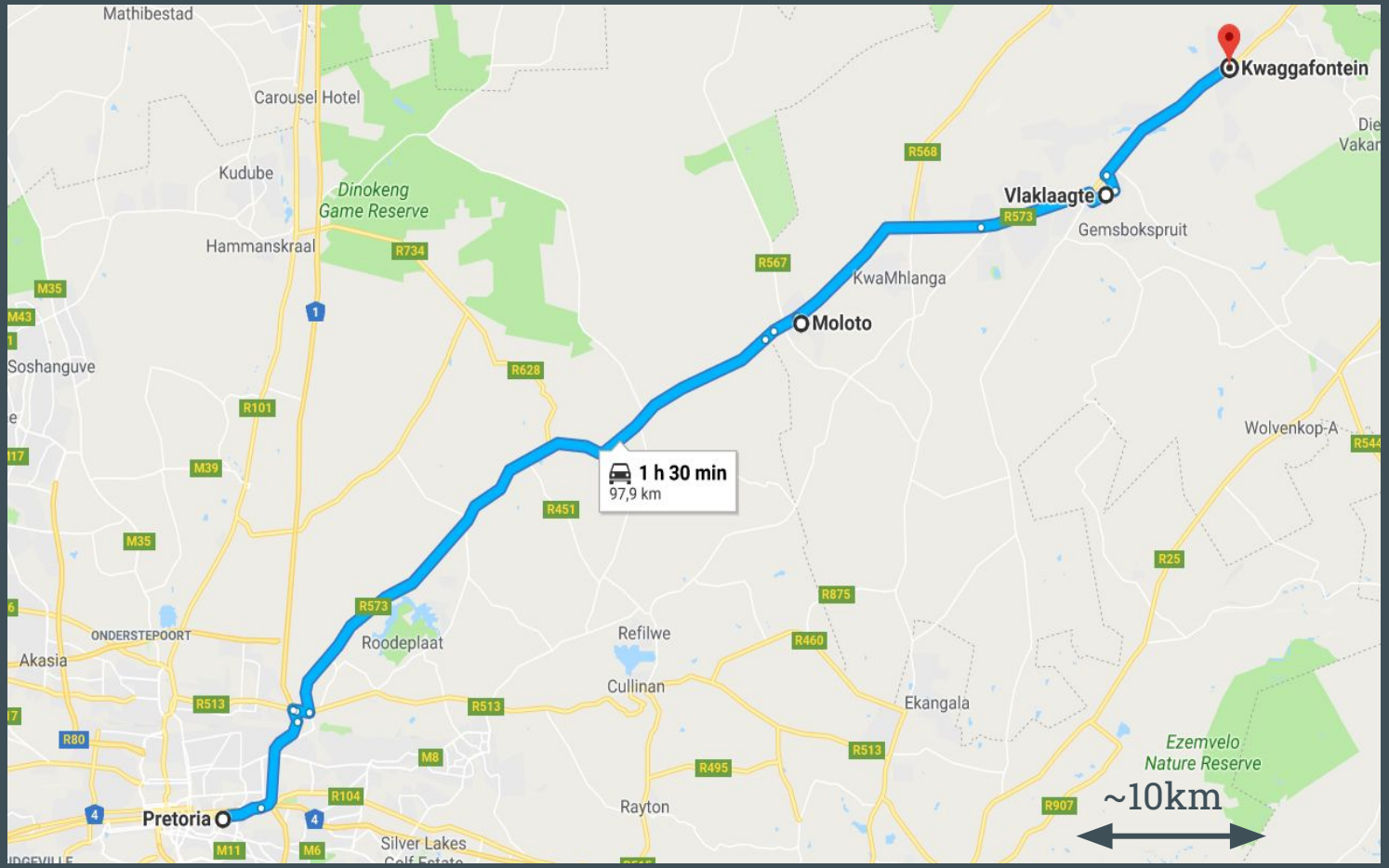


What is the Wi-Fi Aware Protocol

What is the Wi-Fi Aware Protocol

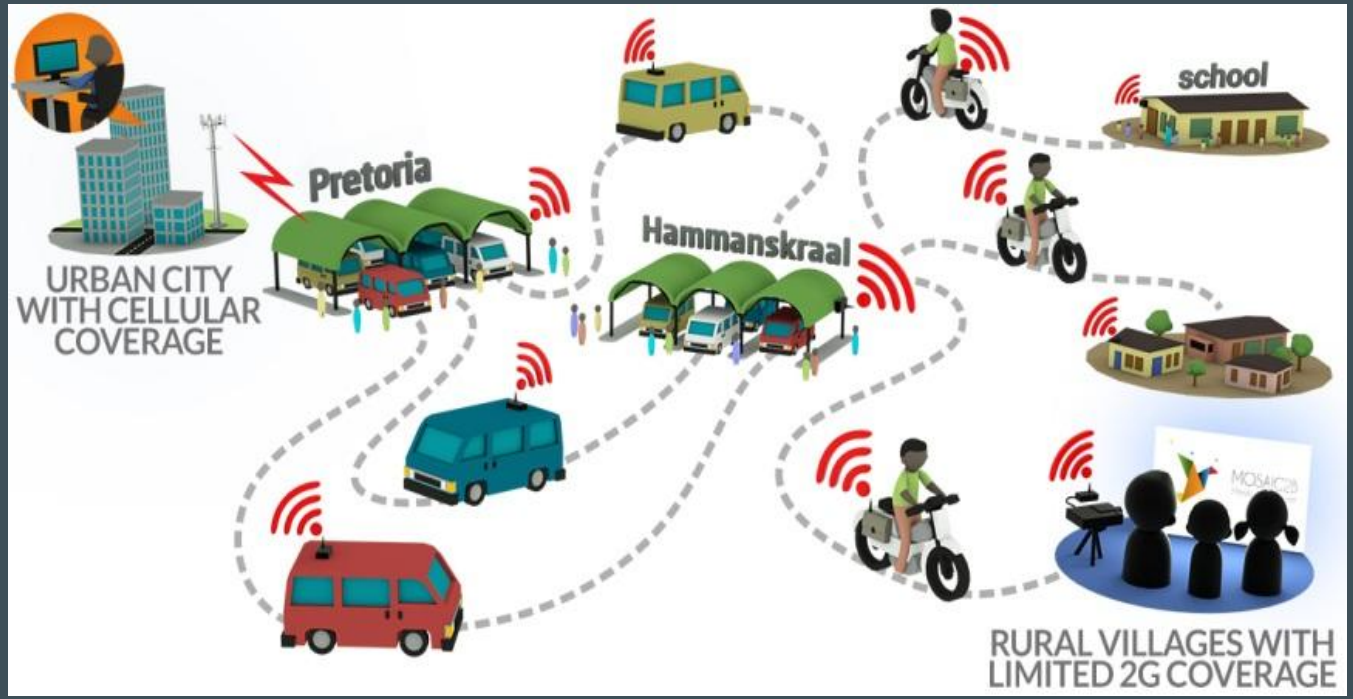
- Allows nearby devices to discover and communicate with each other — without requiring an internet connection or a traditional Wi-Fi access point
- Original use cases: file sharing, multiplayer gaming, smart home, location-based services
- Devices can detect and connect when in physical proximity
- Designed to be energy-efficient, making it suitable for mobile and IoT devices
- Apps discover nearby devices even when not actively running (Android)
- Already supported on Android, just recently iOS, too!

Case 2: Project with Disney, European Commission, Republic South Africa ...


Mathibestad
Carousel Hotel
Kudube
Dinokeng
Game Reserve
Hammanskraal
R734
M35
M43
1
Soshanguve
R101
M39
M35
ONDERSTEPOORT
Akasia
R573
Roodeplaat
R513
R80
M8
R104
Pretoria
M11
M6
Silver Lakes
Refilwe
Cullinan
R513
R495
Rayton
R451
R628
R567
KwaMhlanga
Moloto
R568
Vlaklaagte
R573
Gemsbokspruit
Kwaggafontein
Wolvenkop-A
R25
R875
R460
Ekangala
R513
Ezemvelo
Nature Reserve
R907
~10km
1 h 30 min
97,9 km

Test Case: From Pretoria to Rural South Africa

- Target Villages: Hammanskraal, Moloto, Vlaklaagte, Kwaggafontein



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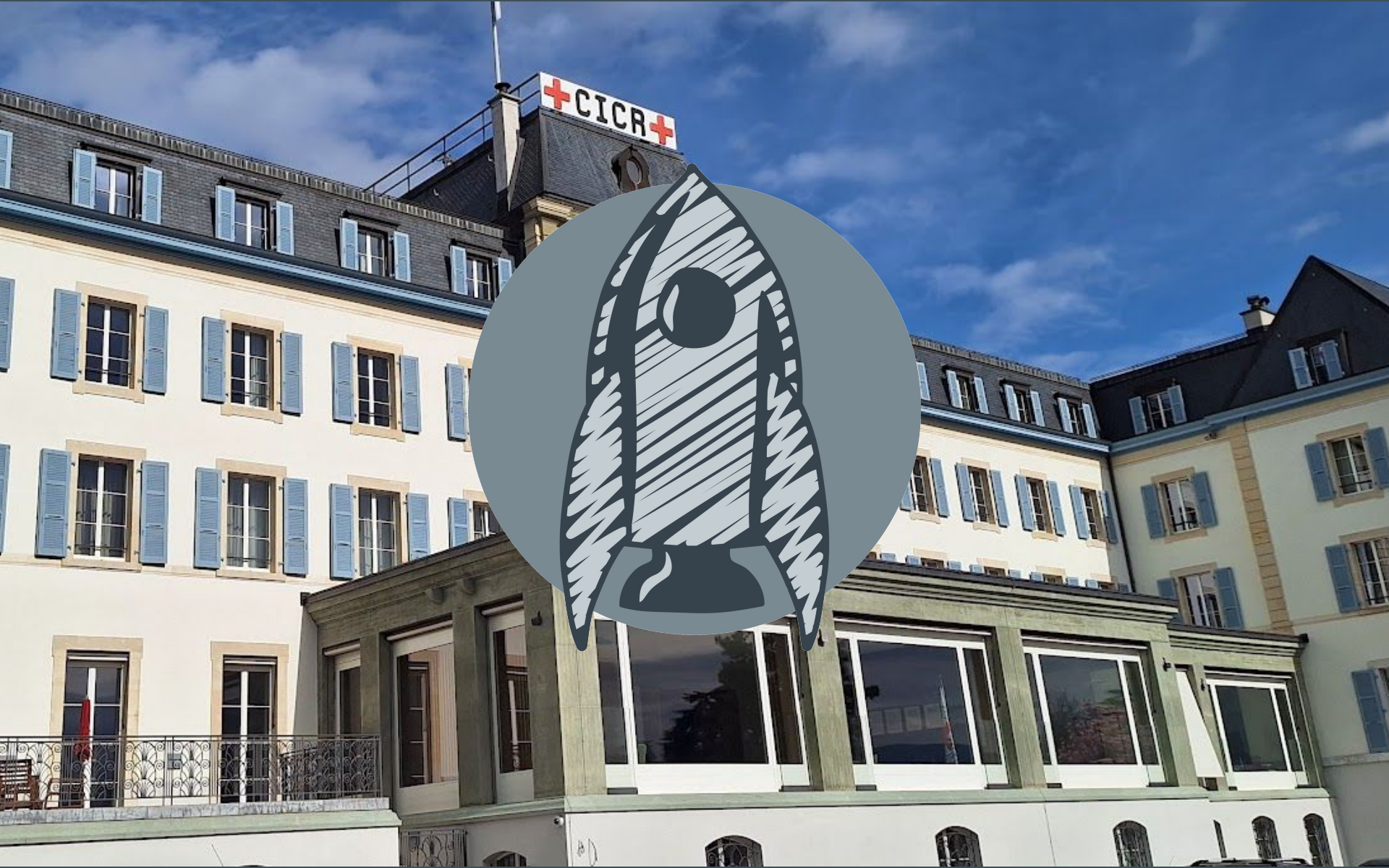
7:22 / 11:06
YouTube

(video)
https://youtu.be/kfgrC-K9L7g?si=BQX7itOyc6Btv92u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